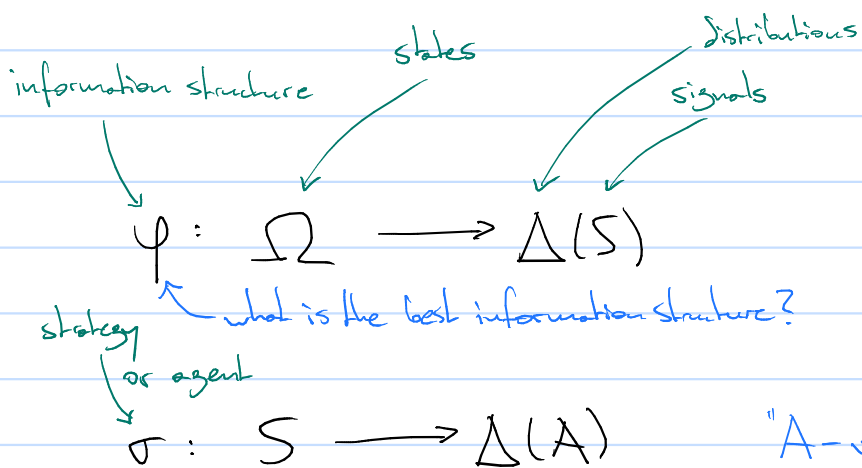


Blackwell's information ordering

All sets are finite



Decisions need to be made in a state $w \in \Omega$ that is observable only through signals.
The payoff of an action $a \in A$ in strategy $w \in \Omega$ is $u(a, w)$.

Note: Any $\alpha: X \rightarrow \Delta(Y)$ extends to $\bar{\alpha}: \Delta(X) \rightarrow \Delta(Y)$ by

$$\pi \mapsto \sum_{x \in X} \pi(x) \alpha(x).$$

Hence, stochastic maps can be composed, by $\beta \circ \alpha := \bar{\beta} \circ \alpha$.

$$\varphi' \leq \varphi \iff \varphi' = \gamma \circ \varphi \text{ for some } \gamma: \Delta(S) \rightarrow \Delta(S')$$

all "strategies" given φ

$$\mathcal{D}(\varphi) = \{ \sigma \circ \varphi \mid \sigma: S \rightarrow \Delta(A) \}$$

$$\sigma \circ \varphi' = \sigma' \circ \varphi$$

Theorem (Blackwell): TFAE

(1) $\varphi' \leq \varphi$;

(2) $\mathcal{D}(\varphi') \subset \mathcal{D}(\varphi)$;

(3) For any payoff $u: A \times \Omega \rightarrow \mathbb{R}$

$$E_{w \in \Omega} (u(\alpha'(w), w)) \leq E_{w \in \Omega} (u(\alpha(w), w))$$

for any Bayesian agent $\sigma: S \rightarrow A$, where $\alpha(w) = \sigma \circ \varphi(w)$.
 i.e. one that optimises $E(u)$ given φ

(1) $\Rightarrow$ (2):

$$\begin{array}{ccc} \Omega & \xrightarrow{\varphi} & S \\ \varphi' \downarrow & \swarrow \gamma & \downarrow \sigma \\ S' & \xrightarrow{\sigma'} & A \end{array} \quad \sigma' = \sigma \circ \gamma$$

(2) $\Rightarrow$ (1):

$$\begin{array}{ccc} \Omega & \xrightarrow{\varphi} & S \\ \varphi' \downarrow & \swarrow \tilde{\sigma} & \downarrow \sigma \\ S' & \xrightarrow[\text{id}_{S'}]{\sigma'} & A = S' \end{array}$$

$\longrightarrow \exists$ by assumption
 $\dashrightarrow \exists$ by conclusion

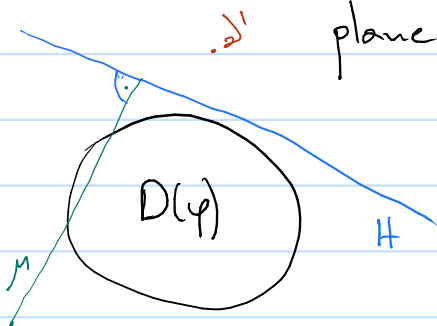
(2) $\Rightarrow$ (3):

$$E(u(\sigma \circ \varphi(w), w)) \geq E(u(\tilde{\sigma} \circ \varphi(w), w)) = E(u(\sigma \circ \varphi'(w), w))$$

true for any $\tilde{\sigma}$, since σ is a Bayesian agent, in particular true for $\tilde{\sigma}$ s.t. $\sigma \circ \varphi' = \tilde{\sigma} \circ \varphi$.

$\neg(2) \Rightarrow \neg(3)$: Let $\alpha' \in \mathcal{D}(\varphi') \setminus \mathcal{D}(\varphi)$. Since $\mathcal{D}(\varphi)$ is compact and convex in $\Delta(A)^\Omega \subset \mathbb{R}^{A \times \Omega}$, it is separated by a hyperplane H from α' . But this means that

$$\mu \cdot \alpha' \geq \mu \cdot H = 0 \geq \mu \cdot \alpha \quad \forall \alpha \in \mathcal{D}(\varphi).$$



equation of hyperplane invariant under the embedding $\Delta(A)^\Omega \subset \mathbb{R}^{A \times \Omega}$

In other words, $E(\mu(\alpha(w), w)) \leq E(\mu(\alpha'(w), w))$,

i.e. the expected utility of the utility function μ is not optimised under σ' .